Jianai Zhao

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PROFESSIONAL PREPARATION

MS, Aerospace Engineering	Technical University of Darmstadt	Germany	Oct 2022 – Dec 2024
BS, Aerospace Engineering Sciences	University of Colorado Boulder	United States	Aug 2017 – May 2022
International Baccalaureate Diploma	Beijing Huijia Private School	China	Sep 2014 – May 2017

AWARD/ FELLOWSHIP/ MEMBERSHIP/ GRANT

Aug 2021 – May 2022	**CROACS Research Grant,** CU Boulder [Smead Aerospace] Senior Design CROACS Research Grant sponsored by [Astroscale U.S.]; managed research budget and procurement.	$5K USD
Apr 2022	**American Institute of Aeronautics and Astronautics Region V,** Student Member [PDF].	Stdnt. Mbr.
Aug 2017	**Outstanding Performance Counselor,** Research Science Initiative (RSI) at Tsinghua.	Awd.
Aug 2015	**RSI Rickoids Alumni,** Research Science Institute (RSI) [Rickoids].	Rickoid

RESEARCH EXPERIENCE

May–Dec 2024	**Master Thesis,** [Chair of Fluid Dynamics]. Title: Neural Network based Shock Detection in Compressible Flow at Space-re-entry Conditions [PDF]. Advisor: [Florian Kummer]. About: Developed AI learning-based artificial viscosity stabilization algorithms from the existing model-based [Persson Sensor], and integrated them into the existing customized CFD ([BoSSS Solver]) for online CFD simulations in high-performance computing (HPC).	Thesis Grade: 1.0 (Highest)
Nov 2022 – Apr 2023	**Master Major Project (GW),** [Institute of Flight Systems and Automatic Control]. Project Topic: Reverse Engineering and Machine Learning for Concept Creation of a Data Mining Hardware [PDF]. Faculty Advisor: [Uwe Klingauf]. About: Fault Tolerant Assessment for Digital Twin Electric Aircraft. Applied 5 different data mining algorithms for fault tolerant assessment from a Very High Frequency digital twin generated by Simulink. Resulted in onboard data logger requirements via data corruption in MATLAB.	Project Grade: 1.3 (2nd-High)
Aug 2021 – May 2022	**Bachelor Capstone Project (GW),** sponsored by Astroscale U.S. [Topic]: Close Range Orbital Attitude Characterization System [PDF]. Advisor: [Yu Takahashi] (NASA Jet Propulsion Laboratory). About: Characterized uncooperative space debris client using a scannerless LiDAR servicer and a customized 3 degree of freedom tumbling rig for ground truth. Employed 3D online template matching for initial pose acquisition and iterative closest point algorithms for dynamic tracking.	Project Grade: 4.0 (Highest)

OUTREACH/ PROFESSIONAL DEVELOPMENT/ SERVICE

Aug 1–4	**Research Science Institute 2026 at Massachusetts Institute of Technology,** [Paper Reader].	SER.
Jul 27–29	**Research Science Institute 2025 at Massachusetts Institute of Technology,** [Paper Reader].	SER.
Jun–Oct 2021	**Smith Lab at CU Boulder,** [Undergrad Research Assistant]. Produced science outreach videos as a part of U.S. National Science Foundation (NSF) funded ([NSF 1553114]) research.	OUT.
Jul–Aug	**Research Science Initiative 2017 at Tsinghua University,** Counselor. [Link]	P.D.
Jul–Aug	**Research Science Initiative 2015 at Tsinghua University,** Aerospace Scholar. [Link].	P.D.